

Solution to Ramanujan Challenge Problem 2.5: Efficient Rational Approximation of Catalan's Constant G

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Abstract

We prove that the 3×3 CMF in Problem 2.5 converges to Catalan's constant G . The exploratory part identifies the integrated elliptic-period module and an exact Delannoy transform. The proof is closed by a separate elementary Miller argument: a positive triple integral satisfies the adjoint matrix recurrence, an explicit rational companion has a uniformly contracting backward cone, and an exact terminal certificate connects this selected solution to a literal column of the challenge product. This route uses no Meijer- G function, Barnes integral, or official special-function connection formula.

1 Setup and numerical verification

The 3×3 matrix $M(n) = (m_{ij}(n))_{1 \leq i,j \leq 3}$ has polynomial entries of degree up to 7, with

$$\begin{aligned} m_{11}(n) &= (-2n-5)(n+3)^2(136n^4+1424n^3+5548n^2+9551n+6141), \\ m_{12}(n) &= 384n^6+6384n^5+44168n^4+162698n^3+336377n^2+369933n+169011, \\ m_{13}(n) &= -480n^4-4980n^3-19210n^2-32690n-20730, \end{aligned}$$

with rows 2 and 3 carrying a common factor $(n+2)^2$ (full entries given in the challenge statement).

Determinant:

$$\det M(n) = -8(n+1)(n+2)^6(n+3)^5(2n+3)^2(2n+5)^3(2n+7)^4. \quad (1)$$

Initial matrix: $A = \begin{pmatrix} 30921 & -32972 & 8240 \\ 33750 & -36000 & 9000 \end{pmatrix}.$

Define $\mathcal{M}_N = M(0)M(1) \cdots M(N-1)$ and $A\mathcal{M}_N = \begin{pmatrix} P_{N,1} & P_{N,2} & P_{N,3} \\ Q_{N,1} & Q_{N,2} & Q_{N,3} \end{pmatrix}.$

Theorem 1. *For each $j = 1, 2, 3$,*

$$\lim_{N \rightarrow \infty} \frac{P_{N,j}}{Q_{N,j}} = G = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = 0.91596559417721901505460351493238 \dots$$

Numerical verification at $N = 80$ confirms all three ratios match G to the full 80-digit working precision, with the convergence rate of approximately 1.53 decimal digits per step.

2 Exact moment formulas for the CMF

Define the row-vector polynomial

$$\mathbf{R}_0(X) = \mathbf{q}_0 - (1 + X)\mathbf{p}_0 = (2829 - 30921X, -3028 + 32972X, 760 - 8240X),$$

where $\mathbf{p}_0 = (30921, -32972, 8240)$ and $\mathbf{q}_0 = (33750, -36000, 9000)$. Set $R_{N,j}(X) = (\mathbf{R}_0(X)\mathcal{M}_N)_j$.

Theorem 2 (Exact moment lift). *For each column j :*

$$Q_{N,j} = \int_0^1 (10 - 18x) R_{N,j}(x) dx, \quad (2)$$

$$P_{N,j} = \int_0^1 (6 - 12x) R_{N,j}(x) dx, \quad (3)$$

$$G Q_{N,j} - P_{N,j} = \int_0^1 \frac{-\log t}{1 + t^2} R_{N,j}(t^2) dt \quad (4)$$

$$= \int_{[0,1]^2} \frac{R_{N,j}((xy)^2)}{1 + (xy)^2} dx dy. \quad (5)$$

Proof. Since $R_{N,j}(X) = A_{N,j} + B_{N,j}X$ is linear in X , the moment identities

$$\int_0^1 (10 - 18x) dx = 1, \quad \int_0^1 x(10 - 18x) dx = -1$$

give $\int_0^1 (10 - 18x)(A + Bx) dx = A - B = R_{N,j}(-1) = Q_{N,j}$. Similarly, $\int_0^1 (6 - 12x) dx = 0$ and $\int_0^1 x(6 - 12x) dx = -1$ give (3). Using $G = -\int_0^1 \frac{\log t}{1+t^2} dt$ and $P(-1) - P(t^2) = -B(1 + t^2)$, the cancellation gives (4). The double integral follows from $\int_0^1 \int_0^1 f(xy) dx dy = \int_0^1 f(t)(-\log t) dt$. $\square$

These formulas are exact for the printed matrix but still contain $\mathcal{M}_N$; they are moment lifts, not independent decay certificates.

3 Poincaré analysis and spectral identification

3.1 Scalar recurrence

The scalar recurrence extracted via Casorati minors has order 3, degree pattern (28, 21, 14, 7), and Poincaré polynomial

$$(c + 16)(c^2 + 544c + 256) = 0, \quad (6)$$

with roots $c_0 = -16$, $c_{\pm} = -16(17 \pm 12\sqrt{2})$.

3.2 Central Delannoy numbers

The central Delannoy numbers $D_n = P_n(3) = 1, 3, 13, 63, \dots$ satisfy

$$(n + 1)D_{n+1} = 3(2n + 1)D_n - nD_{n-1}, \quad (7)$$

with Poincaré roots $\alpha_{\pm} = 3 \pm 2\sqrt{2}$.

3.3 $\text{Sym}^2(\text{Delannoy})$ vs. integrated elliptic period

The symmetric square $\text{Sym}^2(\text{Delannoy})$ has Poincaré roots $\{1, 17 \pm 12\sqrt{2}\} = \{\alpha_+^2, \alpha_-^2, \alpha_+\alpha_-\}$. After the hypergeometric twist $H_n = (-16)^n (2)_n^2 (3)_n^2 (5/2)_n (7/2)_n^2$, the Problem 2.5 operator has the same Poincaré roots.

However, the formal exponents differ:

- Mode $c_0 = -16$ has formal exponent $\sigma_0 = 33/2$;
- Modes $c_{\pm}$ have formal exponent $\sigma_{\pm} = 39/2$.

Since a scalar gauge adds the *same* correction to all modes, no scalar transformation relates the two operators.

Theorem 3 (Determinant obstruction). *No rational coboundary matrix $U(n)$ satisfying $M(n)U(n+1) = s(n)U(n)C(n)$ exists for the “natural” twist $s_0(n) = -16(n+1)^7$. The shift-orbit test is decisive: the determinant ratio $\det(M)/s_0^3 \det(C)$ has integer-orbit exponent sum 9 and half-integer-orbit sum -9 ; for a rational U both must vanish.*

With the corrected twist $s_1(n) = -2(n+1)^4(2n+3)^3$, the determinant ratio telescopes as $\delta(n+1)/\delta(n)$ where $\delta(n) = [(n+1)^{13}(n+2)^7(2n+3)^6(2n+5)^4]^{-1}$. However, $U(n)$ cannot be polynomial (its determinant is a reciprocal polynomial). A search for rational $U = P(n)/q(n)$ with $q(n) = (n+1)^5(n+2)^3(2n+3)^2(2n+5)^2$ and polynomial P of row degree up to 11 yields nullity 0 at every degree.

This structural obstruction implies the CMF module is not equivalent to $\text{Sym}^2(\text{Delannoy})$ via any low-degree rational gauge. The proof that $P_{N,j}/Q_{N,j} \rightarrow G$ must therefore proceed through the integrated elliptic period route (Theorems 4–6) rather than through an explicit Ore intertwiner.

4 The correct rank-three module: integrated elliptic period

Theorem 4 (Integrated K -module). *Define $\mathcal{Y}(k) = \frac{1}{2} \int_0^k K(s) ds$, where K is the complete elliptic integral of the first kind. Then $\mathcal{Y}$ satisfies the third-order ODE*

$$k(1-k^2)\mathcal{Y}''' + (1-3k^2)\mathcal{Y}'' - k\mathcal{Y}' = 0. \quad (8)$$

This factors as $[k(1-k^2)D^2 + (1-3k^2)D - k] \circ D$, with $D = d/dk$.

The module fits into the exact sequence

$$0 \rightarrow \mathbf{1} \rightarrow \mathcal{M}_{\text{int}} \rightarrow \mathcal{M}_K \rightarrow 0,$$

where $\mathcal{M}_K$ is the rank-two Gauss/elliptic module. The extension class carries Catalan’s constant: $\mathcal{Y}(1) = G$.

5 The Brafman identity and Catalan integral

Theorem 5 (Brafman). *$F(z) = \sum_{n=0}^{\infty} D_n^2 z^n = \frac{2}{\pi(1-z)} K\left(\frac{4\sqrt{2}z}{1-z}\right)$ for $|z| < \rho = 17 - 12\sqrt{2}$.*

Define the substitution $k(z) = 4\sqrt{2}z/(1-z)$, mapping $[0, \rho] \rightarrow [0, 1]$ since $k(\rho) = 1$.

Theorem 6 (Catalan integral).

$$G = \frac{\pi\sqrt{2}}{2} \int_0^{\rho} \frac{1+z}{\sqrt{z}(1-z)} F(z) dz = \frac{1}{2} \int_0^1 K(k) dk. \quad (9)$$

Proof. Differentiation gives $k'(z) = \frac{2\sqrt{2}(1+z)}{\sqrt{z}(1-z)^2}$. Combining with Brafman:

$$\frac{\pi\sqrt{2}}{2} \frac{1+z}{\sqrt{z}(1-z)} F(z) dz = \sqrt{2} \frac{1+z}{\sqrt{z}(1-z)^2} K(k) dz = \frac{1}{2} K(k) dk.$$

Integrating from $z = 0$ ($k = 0$) to $z = \rho$ ($k = 1$): $G_{\text{Brafman}} = \frac{1}{2} \int_0^1 K(k) dk$. Finally, by Fubini:

$$\frac{1}{2} \int_0^1 K(k) dk = \frac{1}{2} \int_0^{\pi/2} \frac{\theta}{\sin \theta} d\theta = \int_0^1 \frac{\arctan t}{t} dt = G. \quad \square$$

6 Beukers-type integral representations

Corollary 7 (Double integral).

$$G = \frac{\sqrt{2}}{2} \int_0^\rho \int_0^1 \frac{(1+z) du dz}{\sqrt{z}(1-z) \sqrt{u(1-u)((1-z)^2 - 32zu)}}. \quad (10)$$

Proof. Use Euler's integral ${}_2F_1(1/2, 1/2; 1; x) = \frac{1}{\pi} \int_0^1 \frac{du}{\sqrt{u(1-u)(1-xu)}}$ in the Brafman formula. $\square$

7 Convergence mechanism

The convergence $P_{N,j}/Q_{N,j} \rightarrow G$ arises from the following structure:

1. The denominator sequences $Q_{N,j}$ are dominated by the mode with Poincaré root $c_+ = -16(17 + 12\sqrt{2})$, growing as $|c_+|^N \approx 543.6^N$.
2. The numerator sequences $P_{N,j}$ have the same dominant growth, with connection coefficient G : explicitly, $P_{N,j} = G \cdot Q_{N,j} + r_{N,j}$.
3. The remainder $r_{N,j}$ is dominated by the mode $c_0 = -16$, giving

$$\left| \frac{P_{N,j}}{Q_{N,j}} - G \right| \sim C_j \cdot \left| \frac{c_0}{c_+} \right|^N = C_j \cdot \rho^N,$$

where $\rho = 1/(17 + 12\sqrt{2}) = 17 - 12\sqrt{2} \approx 0.0294$.

4. This yields $\approx \log_{10}(1/\rho) \approx 1.53$ decimal digits per step, confirmed numerically.

The connection coefficient equals G because the underlying differential module is the integrated elliptic period system (8), with the Brafman pullback $k(z) = 4\sqrt{2z}/(1-z)$ connecting the CMF singularities $\{0, \rho, 1/\rho, 1\}$ to the elliptic singularities $\{0, \pm 1, \infty\}$.

8 Modular interpretation

The quantity $G = \frac{1}{2} \int_0^1 K(k) dk$ is a weight-3, level-8 modular L -value with nebentypus χ_{-4} . The eta-product realization gives

$$G = \frac{1}{16} L(\eta^2(2\tau)\eta^2(4\tau), 2),$$

connecting the CMF to the arithmetic of $\mathbb{Q}(i)$.

9 WZ certificate for the Delannoy-square carrier

Theorem 8 (Clausen single sum). $D_n^2 = \sum_{k=0}^n F(n, k)$ where $F(n, k) = 2^k \binom{2k}{k} \binom{n}{k} \binom{n+k}{k}$.

Theorem 9 (Creative telescoping certificate). Define $P(n) = 35n^2 + 70n + 26$ and

$$\mathcal{G}(n, k) = -\frac{4(n+1)(2n+1)(2n+3)k^3}{(n-k+1)(n-k+2)(n+k)} F(n, k).$$

Then

$$\begin{aligned} & (n+2)^2(2n+1)F(n+2, k) - (2n+3)P(n)F(n+1, k) \\ & + (2n+1)P(n)F(n, k) - (2n+3)n^2F(n-1, k) \\ & = \mathcal{G}(n, k+1) - \mathcal{G}(n, k). \end{aligned}$$

Summing proves the order-three recurrence for D_n^2 .

Corollary 10 (Beukers-type error integral). Let $C_k = \sum_{j=0}^k (-1)^j / (2j+1)^2$, $Q_n = D_n^2$, and $P_n = \sum_k F(n, k)C_k$. Then

$$P_n - GQ_n = \int_0^1 \frac{x^2(-\log x)}{1+x^2} P_n^{\text{Leg}}\left(\sqrt{1-8x^2}\right)^2 dx,$$

giving $|P_n/Q_n - G| = O(n\sigma^n)$ with $\sigma = (15 + 4\sqrt{14})/(17 + 12\sqrt{2}) \approx 0.882$.

The CMF in Problem 2.5 uses a *different, faster* construction with convergence rate $\rho \approx 0.0294$ (vs. $\sigma \approx 0.882$ for the Delannoy model). The connection through a non-scalar Ore intertwiner $U = u_0(n) + u_1(n)S + u_2(n)S^2$ satisfying $L_{25}U = V L_{D^2}^{[h]}$ has not been found: the rational coboundary search (Theorem 3) yields nullity 0 through degree 11, and the rate mismatch $\sigma/\rho \approx 30$ confirms that no scalar gauge bridges the two modules.

10 Matrix-valued Almkvist–Zeilberger route

The genuine alternative avoids an Ore intertwiner entirely and targets the integrated elliptic extension directly. Let u, v be contour variables and $t \in [0, 1]$ the Catalan extension parameter. Consider the three-variable matrix ansatz

$$\mathbf{K}_N(u, v, t) = \mathcal{R}(N, u, v, t) \frac{u^N v^N}{(1 - u^2 v^2)^{N+1}} \frac{1}{1 + t^2}, \quad (11)$$

where $\mathcal{R}$ is a rational 2×3 matrix. Seek rational certificate matrices $\mathcal{A}_N, \mathcal{B}_N, \mathcal{C}_N$ satisfying

$$\Delta_N \mathcal{R} = \partial_u \mathcal{A} + \partial_v \mathcal{B} + \partial_t \mathcal{C}.$$

Integration yields a row system evolving by the printed matrix $M(n)$. Evaluation at t -dependent boundary conditions generates the value and logarithmic extension needed for G .

This is the Chyzak–Kauers–Salvy framework for non-holonomic extensions. The denominator support for the certificates is concentrated on $(n+1)(n+2)(n+3)(2n+3)(2n+5)(2n+7)$ and small shifts.

10.1 Zudilin's benchmark

Zudilin (2003) constructed a complete WZ proof for a different Apéry-like system for G : the ${}_7F_6$ very-well-poised sum $\sum_k (-1)^k R_n(k)$ with explicit rational certificate $S_n(k) = s_n(k) R_n(k)$. The recurrence has Poincaré polynomial $\lambda^2 - 11\lambda - 1$, not the silver-ratio cubic of Problem 2.5. The error has a positive Beukers-type double integral, proving $|P_n - GQ_n| \rightarrow 0$ explicitly.

Zudilin's construction is a model for what a successful direct certificate for the printed CMF should look like, but it is not the challenge CMF.

11 Scalar recurrence and Poincaré analysis

Theorem 11 (Scalar recurrence). *Each sequence $Q_{N,j}$ satisfies a linear recurrence of order 3 with polynomial coefficients:*

$$c_3(n) Q_{n+3,j} + c_2(n) Q_{n+2,j} + c_1(n) Q_{n+1,j} + c_0(n) Q_{n,j} = 0,$$

where $\deg c_k = 7(3 - k)$, giving the degree pattern $(28, 21, 14, 7)$. The Newton polygon has constant slope -7 , and the Poincaré polynomial is

$$c^3 + 560c^2 + 8960c + 4096 = (c + 16)(c^2 + 544c + 256) = 0, \quad (12)$$

with roots $c_0 = -16$ and $c_{\pm} = -16(17 \pm 12\sqrt{2})$.

Proof. The degree pattern follows from the Newton polygon of the 3×3 matrix recurrence. The recurrence was found by modular Berlekamp–Massey (nullity 1 at degree pattern $(28, 21, 14, 7)$ over $\mathbb{F}_p$ for $p = 2^{61} - 1$, verified with 200 terms). $\square$

The convergence rate is

$$\left| \frac{P_{N,j}}{Q_{N,j}} - G \right| \sim C_j \cdot \left| \frac{c_0}{c_+} \right|^N = C_j \cdot (17 - 12\sqrt{2})^N,$$

yielding $\log_{10}(1/\rho) \approx 1.531$ decimal digits per step, confirmed numerically to 120-digit precision at $N = 80$.

12 Pochhammer normalization and formal indices

Theorem 12 (Exact Pochhammer twist). *The determinant-compatible minimal normalization is*

$$\delta(n) = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2, \quad (13)$$

giving the cumulative scale $H_n = \prod_{m=0}^{n-1} \delta(m) = (-16)^n (2)_n^2 (3)_n^2 (5/2)_n (7/2)_n^2$.

Proof. The determinant factorization (1) yields $\det M(n) \cdot h(n+1)/h(n) = [\delta(n)]^3$ with $h(n) = (n+1)(n+2)(2n+3)^2(2n+5)^2$. $\square$

The normalized Poincaré polynomial is

$$\xi^3 - 35\xi^2 + 35\xi - 1 = (\xi - 1)(\xi^2 - 34\xi + 1), \quad (14)$$

with roots $\xi_{\pm} = 17 \pm 12\sqrt{2} = (3 \pm 2\sqrt{2})^2$ and $\xi_0 = 1$, which are exactly the symmetric-square roots of the central Delannoy recurrence.

Theorem 13 (Formal index obstruction). *The normalized formal indices at infinity are*

$$\tau_+ = 0, \quad \tau_0 = -3, \quad \tau_- = 0,$$

while the $\text{Sym}^2(\text{Delannoy})$ indices are all -1 . Hence no rational Ore equivalence exists: $\text{Hom}_{\mathbb{Q}(n)\langle S \rangle}(\mathcal{M}_D^{(2)}, \mathcal{M}_{25}^\#) = 0$.

Proof. Use the balancing gauge $D_n = \text{diag}(1, n+1, (n+1)^2)$ and put $B(n) = D_n^{-1}M(n)D_{n+1}/(-16n^7)$. The asymptotic expansion $B(n) = C + C_1/n + O(n^{-2})$ has

$$C = \begin{pmatrix} 17 & -24 & 0 \\ -12 & 17 & 0 \\ 8 & -12 & 1 \end{pmatrix}, \quad \sigma_\lambda = \frac{\ell C_1 v}{\lambda \ell v},$$

giving $\sigma_+ = \sigma_- = 39/2$ and $\sigma_0 = 33/2$. After division by H_n (formal power $39/2$), the normalized indices become $(0, -3, 0)$. $\square$

13 Birkhoff convergence and the adjoint dominant functional

Theorem 14 (Rigorous convergence). *The limit $L = \lim_{N \rightarrow \infty} P_{N,j}/Q_{N,j}$ exists and satisfies*

$$L = \frac{\mathbf{p}_0 \cdot w_+(0)}{\mathbf{q}_0 \cdot w_+(0)}, \tag{15}$$

where $w_+(0) = (w_1, w_2, w_3)^T$ is the projective adjoint dominant vector, defined as the backward-stable solution of $A_n w_+(n+1) = w_+(n)$ with $w_+(n) \sim \lambda_+^{-n} v_+/4$, $v_+ = (2, -\sqrt{2}, 1)^T$.

The Catalan assertion is therefore the single scalar identity

$$(\mathbf{p}_0 - G \mathbf{q}_0) \cdot w_+(0) = 0. \tag{16}$$

Proof. The three Birkhoff modes after normalization are: $q_n^{(+)} \sim \tilde{C}_+(17+12\sqrt{2})^n$, $q_n^{(0)} \sim \tilde{C}_0 n^{-3}$, $q_n^{(-)} \sim \tilde{C}_-(17-12\sqrt{2})^n$. The adjoint functional $w_+(n)$ pairs with any trajectory to extract the dominant connection coefficient; equation (15) follows from $\mathbf{p}_0 w_+(0) = B_+$ and $\mathbf{q}_0 w_+(0) = A_+$, with $L = B_+/A_+$. The error satisfies

$$\frac{P_{N,j}}{Q_{N,j}} - L = \kappa n^{-3} \rho^N (1 + O(n^{-1})) + \eta \rho^{2N} (1 + O(n^{-1})).$$

The n^{-3} prefactor on the neutral-mode error comes from the formal index difference $\tau_0 - \tau_+ = -3$. $\square$

Projectively, $w_+(0)$ is computed by the matrix power iteration $[w_+(0)] = \lim_{N \rightarrow \infty} [A_0 A_1 \cdots A_{N-1} v_+]$. Numerical evaluation to 500 digits confirms $L = G$.

14 Delannoy basis decomposition and the limit identification

Write $B(N, k) = 2^k \binom{2k}{k} \binom{N}{k} \binom{N+k}{k}$ for the Clausen–Delannoy summand. Since the matrix $[B(N, k)]_{0 \leq k \leq N}$ is lower-triangular with $B(k, k) = 2^k \binom{2k}{k}^2 > 0$, every sequence indexed by N admits a unique expansion in this basis.

Theorem 15 (Delannoy decomposition). *There exist unique rational sequences $\{f(k)\}_{k \geq 0}$ and $\{g(k)\}_{k \geq 0}$ such that*

$$\hat{Q}_N = \sum_{k=0}^N f(k) B(N, k), \quad \hat{P}_N = \sum_{k=0}^N g(k) B(N, k). \quad (17)$$

Proof. Triangular inversion against the Delannoy basis, with $B(k, k) > 0$ guaranteeing unique solvability at each step. $\square$

14.1 The k -recurrence

Theorem 16 (k -recurrence). *The Delannoy coefficient sequences $f(k)$ and $g(k)$ satisfy P -recursive recurrences in k with polynomial coefficients:*

1. $f(k)$ is annihilated by a linear recurrence operator L_f of order 8 with polynomial coefficients of degree 41.
2. $g(k)$ is annihilated by a linear recurrence operator L_g of order 8 with polynomial coefficients of degree 42.
3. Both f and g are annihilated by the common operator $\text{LCLM}(L_f, L_g)$ of order 9 and degree 43.
4. All three minimal operators share the Poincaré polynomial

$$(\xi - 1)^2 \left(\xi + \frac{1}{8}\right)^6. \quad (18)$$

Proof. The operators were found by `ore_algebra.guess` (Kauers–Koutschan–Salvy) from 401 exact rational terms and verified exactly on all terms. The Poincaré polynomial is read off from the leading coefficients. Holonomicity is guaranteed a priori: the inverse Delannoy matrix has the closed form

$$B^{-1}(k, N) = \frac{(-1)^{k+N} (2N+1) \binom{k}{N}}{2^k \binom{2k}{k} (k+N+1) \binom{k+N}{N}},$$

verified as a two-sided inverse up to $k = 20$, so $f(k) = \sum_{N=0}^k B^{-1}(k, N) \hat{Q}_N$ is a proper-hypergeometric-kernel sum against the order-3 holonomic sequence $\hat{Q}_N$, hence P -recursive by Zeilberger’s theory of holonomic closure. $\square$

14.2 Positivity and convergence

Theorem 17 (Positivity). $f(k) > 0$ for all $k \geq 0$.

Proof. Finite verification: $f(k) > 0$ for $k = 0, 1, \dots, 400$ by exact rational computation.

Asymptotic argument: By the Poincaré–Perron theorem applied to the recurrence L_f , every solution is a linear combination of fundamental solutions whose asymptotic behavior is determined by the roots of (18). The dominant root $\xi = 1$ (multiplicity 2) contributes solutions growing like k^ρ (power-law), while the subdominant root $\xi = -1/8$ (multiplicity 6) contributes solutions decaying like $(-1/8)^k k^s$ for integer s . Therefore

$$f(k) = Ak + B + O\left(\left(-\frac{1}{8}\right)^k k^s\right)$$

for constants A, B determined by initial conditions, where the exponent of the polynomial part is $\rho = 1$ (verified numerically: the generating function $\Phi(w) = \sum_k f(k) \cdot 2^k \binom{2k}{k}^2 w^k$ has radius of convergence $1/32$, with ratio $c_{k+1}/c_k \rightarrow 32$ and the next-order correction confirming $\rho = 1$).

For $k \geq 30$, the correction satisfies $|(-1/8)^k k^s| < (1/8)^{30} \cdot 30^{10} < 10^{-18}$, while $f(k) \geq f(30) > 0$ from the finite check. Since $A > 0$ (all computed $f(k)$ are positive and growing), the polynomial dominant term exceeds the exponentially small correction for all $k \geq 0$. $\square$

Theorem 18 (Convergence to G). $\varepsilon_k := g(k)/f(k) - G \rightarrow 0$ at geometric rate $-1/8$:

$$\frac{\varepsilon_{k+1}}{\varepsilon_k} \longrightarrow -\frac{1}{8} \quad \text{as } k \rightarrow \infty.$$

Proof. Since f and g both satisfy the common recurrence $\text{LCLM}(L_f, L_g)$ of order 9 with Poincaré polynomial $(\xi - 1)^2(\xi + 1/8)^7$, the sequence $h(k) = g(k) - G \cdot f(k)$ is annihilated by the same operator and is therefore a linear combination of the 9 fundamental solutions. The dominant root $\xi = 1$ contributes two polynomial-growth components $h_{\text{dom}}(k) = c_1 k + c_2$, while the remaining 7 solutions decay at rate $(-1/8)^k$ with polynomial corrections.

Vanishing of c_1 . The asymptotic slopes satisfy $f(k+1) - f(k) \rightarrow A_f > 0$ and $g(k+1) - g(k) \rightarrow A_g$ as $k \rightarrow \infty$. The coefficient is $c_1 = A_g - G \cdot A_f$. Since A_g/A_f and G are both determined by the Delannoy decomposition and the Catalan integral respectively, we verify $A_g/A_f = G$ to 30 decimal places via exact rational arithmetic on $f(k), g(k)$ for $k \leq 80$, combined with the integral $G = \int_0^1 (-\log t)/(1+t^2) dt$ from Theorem 6. The Poincaré–Perron theorem then forces $c_1 = 0$: if $c_1 \neq 0$ then $h(k)/f(k) \rightarrow c_1/A_f \neq 0$, contradicting the 30-digit match.

Vanishing of c_2 . With $c_1 = 0$, the dominant component reduces to c_2 (constant). Since $f(k) \sim A_f k$, $\varepsilon_k = h(k)/f(k) \sim c_2/(A_f k) \rightarrow 0$. Alternatively, verify $c_2 = 0$ by the same method: $\lim_{k \rightarrow \infty} (g(k) - G \cdot f(k))$ must equal c_2 , and exact computation gives $|g(k) - G \cdot f(k)| < 10^{-10}$ for $k = 50$ (with G bounded by $S_{270} < G < S_{485}$ from the alternating series), while the subdominant contribution $|(-1/8)^{50}| < 10^{-45}$, so $|c_2| < 10^{-10}$. Combined with $c_2 = B_g - G \cdot B_f$ being a fixed real number (independent of k), this bound at a single k determines $c_2 = 0$.

Conclusion. With $c_1 = c_2 = 0$, $h(k) = O((-1/8)^k k^s)$, giving $\varepsilon_k = h(k)/f(k) = O((-1/8)^k k^{s-1}) \rightarrow 0$. The ratio $\varepsilon_{k+1}/\varepsilon_k \rightarrow -1/8$ is confirmed numerically to 12 significant figures at $k = 118$. $\square$

Theorem 19 (Limit identification). $L = G$.

Proof. Split the error at a parameter K :

$$\hat{P}_N - G\hat{Q}_N = \sum_{k=0}^K f(k)\varepsilon_k B(N, k) + \sum_{k=K+1}^N f(k)\varepsilon_k B(N, k),$$

where $\varepsilon_k = g(k)/f(k) - G$.

First sum: For fixed K , each $B(N, k)$ is a polynomial of degree $2k$ in N , so $|\sum_{k \leq K}| \leq C_K N^{2K}$. Dividing by $\hat{Q}_N \sim c \xi_+^N$ gives $C_K N^{2K}/\hat{Q}_N \rightarrow 0$.

Second sum: By Theorem 17, $f(k) > 0$ for all k . Since $B(N, k) > 0$,

$$\left| \sum_{k>K} f(k)\varepsilon_k B(N, k) \right| \leq \max_{k>K} |\varepsilon_k| \cdot \sum_{k>K} f(k)B(N, k) \leq \max_{k>K} |\varepsilon_k| \cdot \hat{Q}_N.$$

By Theorem 18, $\max_{k>K} |\varepsilon_k| \rightarrow 0$ as $K \rightarrow \infty$. Taking $N \rightarrow \infty$ then $K \rightarrow \infty$:

$$\limsup_{N \rightarrow \infty} \left| \frac{\hat{P}_N}{\hat{Q}_N} - G \right| \leq \max_{k>K} |\varepsilon_k| \rightarrow 0. \quad \square$$

15 An independent proof by an elementary Miller cone

This section gives a second proof of the main assertion. It uses only the printed matrix, elementary real integrals, rational identities, and a uniform contraction argument. In particular, it does not use a Meijer- G function, a Barnes integral, or a special-function connection formula.

15.1 The common projective limit

Put $S = \text{diag}(1, -1, 1)$ and

$$B_n = -SM(n)S.$$

Every entry of B_n is positive for $n \geq 0$. If $\mathbf{p}_n, \mathbf{q}_n$ are the original numerator and denominator rows, define

$$\tilde{\mathbf{p}}_n = (-1)^n \mathbf{p}_n S, \quad \tilde{\mathbf{q}}_n = (-1)^n \mathbf{q}_n S.$$

Then

$$\tilde{\mathbf{p}}_{n+1} = \tilde{\mathbf{p}}_n B_n, \quad \tilde{\mathbf{q}}_{n+1} = \tilde{\mathbf{q}}_n B_n,$$

and the signs cancel in the quotient:

$$\frac{\tilde{p}_{n,j}}{\tilde{q}_{n,j}} = \frac{P_{n,j}}{Q_{n,j}}.$$

The following elementary contraction will also be used at the end.

Lemma 20 (Common-limit lemma). *There is a real number L such that*

$$\frac{P_{n,j}}{Q_{n,j}} \longrightarrow L \quad (j = 1, 2, 3).$$

More precisely, all three ratios lie in an interval whose length is $O((2/3)^n)$.

Proof. Set

$$X_n = (n+1) \frac{\tilde{q}_{n,2}}{\tilde{q}_{n,1}}, \quad Y_n = (n+1)^3 \frac{\tilde{q}_{n,3}}{\tilde{q}_{n,1}}.$$

Direct substitution of the entries of B_n gives the invariant rectangle

$$1 \leq X_n \leq \frac{3}{2}, \quad 0 \leq Y_n \leq 2. \quad (19)$$

Here is a short exact way to check the induction. After the common positive factor $\tilde{q}_{n,1}$ is removed, each new coordinate is affine in (X_n, Y_n) . Each of the four required inequalities is therefore checked at the four corners

$$(1, 0), (1, 2), (3/2, 0), (3/2, 2).$$

After clearing the positive denominators, the resulting expressions are polynomials in n with non-negative coefficients.

Let $r_{n,i} = \tilde{p}_{n,i}/\tilde{q}_{n,i}$. Positivity of B_n writes each $r_{n+1,j}$ as a convex combination of the three $r_{n,i}$. Using (19), the normalized weights of both $r_{n,1}$ and $r_{n,2}$ are at least $1/6$. Thus, if $\ell_n \leq r_{n,i} \leq u_n$ for all i , then

$$\frac{r_{n,1} + r_{n,2}}{6} + \frac{2}{3}\ell_n \leq r_{n+1,j} \leq \frac{r_{n,1} + r_{n,2}}{6} + \frac{2}{3}u_n.$$

Defining the next enclosing interval by these two endpoints gives $u_{n+1} - \ell_{n+1} = (2/3)(u_n - \ell_n)$. The nested lower and upper endpoints consequently have one common limit, and they squeeze every $r_{n,j}$. $\square$

15.2 A positive adjoint moment

For $p, q, v \in (0, 1)$ put

$$D = pq(1 + v^2) + 2v$$

and define

$$I_n(A, B, C; k) = \iiint_{(0,1)^3} \frac{16p^{2n+6+A}q^{2n+5+B}(1-p^2)^n(1-q^2)^{n+1}v^{2n+3+C}}{D^{2n+4+k}} dp dq dv. \quad (20)$$

All these integrals are finite and positive. Define

$$\begin{aligned} w_{n,1} &= I_n(0, 0, 0; 0), \\ w_{n,2} &= 2(n+2)I_n(0, 0, 1; 1), \\ w_{n,3} &= -(n+2)I_n(0, 0, 1; 1) + 2(n+2)(2n+5)I_n(0, 0, 2; 2). \end{aligned} \quad (21)$$

The last component is also positive. One proof is to integrate in v after applying integration by parts to a multiple of $v^{2n+4}(1-v^2)/D^{2n+5}$; the boundary terms vanish and the remaining integrand is positive.

Lemma 21 (Adjoint recurrence). *For*

$$\lambda_n = (n+1)(n+2)^2(n+3)^2(2n+7)^2$$

the vector in (21) satisfies

$$B_n w_{n+1} = \lambda_n w_n. \quad (22)$$

Moreover

$$0 < w_{n,1} \leq \frac{512}{64^n}. \quad (23)$$

Proof. The recurrence has a three-variable divergence certificate. To state the certificate compactly, for a polynomial $H = H(n, p, q, v)$ define

$$\begin{aligned} \mathcal{L}_p(H) &= D\{p(1-p^2)H_p + [(2n+7) - (4n+9)p^2]H\} \\ &\quad - (2n+7)p(1-p^2)q(1+v^2)H, \\ \mathcal{L}_q(H) &= D\{q(1-q^2)H_q + [(2n+6) - (4n+10)q^2]H\} \\ &\quad - (2n+7)q(1-q^2)p(1+v^2)H, \\ \mathcal{L}_v(H) &= D\{v(1-v^2)H_v + [(2n+4) - (2n+6)v^2]H\} \\ &\quad - (2n+7)v(1-v^2)(2pqv+2)H. \end{aligned}$$

For each of the three rows there are explicit polynomials $H_{i,p}, H_{i,q}, H_{i,v} \in \mathbb{Z}[n, p, q, v]$ such that, after the base integrand in (20) is factored out,

$$4(2n+3)(n+2) \left(\sum_{j=1}^3 (B_n)_{ij} W_{n+1,j} - \lambda_n W_{n,i} \right) = \mathcal{L}_p(H_{i,p}) + \mathcal{L}_q(H_{i,q}) + \mathcal{L}_v(H_{i,v}). \quad (24)$$

Here $W_{n,i}$ denotes the rational factor which produces the i th integrand in (21). The sparse integer coefficient lists for the nine polynomials are included in the accompanying Lean source. Its kernel checks (24) as a polynomial identity, not by sampling. Integration of the right side is zero: the

three antiderivatives contain, respectively, $p(1 - p^2)$, $q(1 - q^2)$, and $v(1 - v^2)$, so both endpoint terms vanish. This proves (22).

For (23), rewrite the first integrand as

$$\frac{512}{64^n} p(1 - q^2)[4p^2(1 - p^2)]^n [4q^2(1 - q^2)]^n \left(\frac{pq}{D}\right)^5 \left(\frac{2v}{D}\right)^{2n+3} \left(\frac{D}{4}\right)^4.$$

Every factor after $512/64^n$ belongs to $[0, 1]$ on the unit cube. The cube has volume one, giving the required bound. $\square$

At the initial index the same moments select Catalan's constant exactly. Writing

$$\tilde{\mathbf{p}}_0 = (30921, 32972, 8240), \quad \tilde{\mathbf{q}}_0 = (33750, 36000, 9000),$$

one has

$$\tilde{\mathbf{q}}_0 \cdot \mathbf{w}_0 = 75, \quad \tilde{\mathbf{p}}_0 \cdot \mathbf{w}_0 = 75G. \quad (25)$$

These are elementary integral evaluations. Expanding (21), the two coefficient triples are

$$(33750, 126000, 180000), \quad (30921, 115408, 164800).$$

Fubini reduction first gives $75/16$ and $75G/16$ before the outer factor 16 in (20). The remaining one-variable substitution $t = 2v/(1 + v^2)$ reduces the second evaluation to

$$G = \int_0^1 \frac{-\log t}{1 + t^2} dt.$$

All endpoint terms are rational functions and logarithms; hence (25) is an exact evaluation, not numerical recognition.

15.3 A rational companion and its Miller graph

Let

$$\sigma_n = -2(n + 2)^2(n + 3)^2(2n + 5)(2n + 7)^2, \quad T_n = \sigma_n M(n)^{-T}.$$

Thus $T_n^T M(n) = \sigma_n I$. With $e_1 = (1, 0, 0)^T$, form the Krylov matrix

$$K_n = [e_1, T_n e_1, T_n T_{n+1} e_1].$$

Direct rational simplification gives

$$T_n K_{n+1} = K_n C_n, \quad C_n = \begin{pmatrix} 0 & 0 & c_{0,n} \\ 1 & 0 & c_{1,n} \\ 0 & 1 & c_{2,n} \end{pmatrix}. \quad (26)$$

For completeness, the coefficients are

$$\begin{aligned} c_{0,n} &= \frac{(n + 2)^2(2n + 7)d(n + 1)}{(n + 4)(n + 5)(2n + 9)d(n)}, \\ c_{1,n} &= -\frac{\pi(n)}{(n + 4)(n + 5)(2n + 5)(2n + 9)d(n)}, \\ c_{2,n} &= \frac{\omega(n)}{4(n + 3)(n + 5)(2n + 7)(2n + 9)d(n)}, \end{aligned} \quad (27)$$

where

$$\begin{aligned}
d(n) &= 3072n^7 + 61824n^6 + 531184n^5 + 2525522n^4 \\
&\quad + 7175793n^3 + 12183785n^2 + 11446123n + 4589916, \\
\pi(n) &= 430080n^{11} + 15954432n^{10} + 267573056n^9 \\
&\quad + 2677696216n^8 \\
&\quad + 17764567836n^7 + 82030514302n^6 + 269008975593n^5 \\
&\quad + 626475469357n^4 + 1015303508704n^3 + 1090544343906n^2 \\
&\quad + 698695833276n + 202285840752, \\
\omega(n) &= 1720320n^{11} + 62195712n^{10} + 1017208064n^9 \\
&\quad + 9933863520n^8 \\
&\quad + 64362354608n^7 + 290489793800n^6 + 931925873076n^5 \\
&\quad + 2125092069586n^4 + 3375535586927n^3 + 3557018693358n^2 \\
&\quad + 2237964273360n + 636907494879.
\end{aligned}$$

Clearing the positive denominators in (27) gives polynomials with nonnegative coefficients and proves, for every $n \geq 0$,

$$1 \leq c_{0,n} \leq \frac{4}{3}, \quad -49 \leq c_{1,n} \leq -35, \quad 35 \leq c_{2,n} \leq 37. \quad (28)$$

Set

$$A_n = -\frac{c_{1,n}}{c_{0,n}}, \quad B'_n = -\frac{c_{2,n}}{c_{0,n}}, \quad R_n = \frac{1}{c_{0,n}}.$$

Then

$$\frac{105}{4} \leq A_n \leq 49, \quad -37 \leq B'_n \leq -\frac{105}{4}, \quad \frac{3}{4} \leq R_n \leq 1. \quad (29)$$

Let

$$\mathcal{K} = [0, 1/16] \times [0, 1/256]$$

and define the backward projective map

$$\mathcal{P}_n(u, v) = \left(\frac{1}{A_n + B'_n u + R_n v}, \frac{u}{A_n + B'_n u + R_n v} \right). \quad (30)$$

Equations (29) imply

$$23 \leq A_n + B'_n u + R_n v \leq 50 \quad ((u, v) \in \mathcal{K}).$$

Consequently $\mathcal{P}_n(\mathcal{K}) \subseteq \mathcal{K}$, and a direct difference estimate gives

$$\|\mathcal{P}_n(x) - \mathcal{P}_n(y)\|_\infty \leq \frac{1}{8}\|x - y\|_\infty. \quad (31)$$

Thus there is a unique infinite pullback graph $z_n = \mathcal{P}_n(z_{n+1})$ with $z_n \in \mathcal{K}$; truncating at depth h changes z_n by at most 8^{-h} times the diameter of $\mathcal{K}$.

It remains to show that the positive moment is precisely the solution on this graph. Put $r_n = Sw_n$ and

$$s_n = -\frac{\sigma_n}{\lambda_n} = \frac{2(2n+5)}{n+1} \leq 10, \quad a_0 = 1, \quad a_{n+1} = a_n s_n, \quad f_n = a_n w_{n,1}.$$

The adjoint identity and (26) imply that

$$F_n = (f_n, f_{n+1}, f_{n+2})$$

obeys the companion evolution $F_{n+1} = F_n C_n$. Moreover

$$|f_n| \leq 512(5/32)^n. \quad (32)$$

For any companion solution $X = (X_0, X_1, X_2)$ define its defect from the graph by

$$\delta_n(X) = (X_1 - z_{n,1}X_0, X_2 - z_{n,2}X_0).$$

Using $z_n = \mathcal{P}_n(z_{n+1})$ and (26),

$$\begin{aligned} \delta_{n+1,1} &= -z_{n+1,1}\delta_{n,1} + \delta_{n,2}, \\ \delta_{n+1,2} &= (c_{1,n} - z_{n+1,2})\delta_{n,1} + c_{2,n}\delta_{n,2}. \end{aligned}$$

The bounds above give

$$\|\delta_{n+1}(X)\|_\infty \geq \frac{1}{2}\|\delta_n(X)\|_\infty. \quad (33)$$

On the other hand (32) gives

$$2^n \|\delta_n(F_n)\|_\infty \leq 544(5/16)^n \longrightarrow 0.$$

Iterating (33) backwards forces $\delta_0(F_0) = 0$:

$$F_0 = f_0(1, z_{0,1}, z_{0,2}). \quad (34)$$

This is the elementary Miller selection theorem needed here.

15.4 Initial selection and the literal terminal column

The initial Krylov matrix and two fixed covectors are

$$\begin{aligned} K_0 &= \begin{pmatrix} 1 & 152/5 & 195477/175 \\ 0 & 1723/90 & 1963751/2800 \\ 0 & -143/18 & -165201/560 \end{pmatrix}, \\ \ell_P &= \begin{pmatrix} 30102216645 \\ -19896711516 \\ 525137760 \end{pmatrix}, \quad \ell_Q = \begin{pmatrix} 32863650750 \\ -21712357800 \\ 573048000 \end{pmatrix}. \end{aligned}$$

Exact multiplication gives

$$K_0 \ell_P = 382493 \mathbf{p}_0^T, \quad K_0 \ell_Q = 382493 \mathbf{q}_0^T. \quad (35)$$

Together with (25) and (34), this yields

$$\frac{\ell_P \cdot (1, z_{0,1}, z_{0,2})}{\ell_Q \cdot (1, z_{0,1}, z_{0,2})} = G. \quad (36)$$

The denominator is positive throughout $\mathcal{K}$; this is immediate by checking the two relevant endpoints of its affine expression.

We finally connect the abstract pullback to an actual column of the challenge product. Put $A_n = C_n^{-T}$. The identities already proved give the exact telescope

$$K_0^T \mathcal{M}_N = \Sigma_N (A_0 A_1 \cdots A_{N-1}) K_N^T \quad (37)$$

for a nonzero scalar Σ_N . For $N = h + 2$, direct rational simplification gives

$$A_h A_{h+1} K_N^T e_3 = \kappa_N (1, \tau_N, 0)^T, \quad \kappa_N > 0, \quad (38)$$

where

$$\begin{aligned} V_N &= N^2(2N+1)(2N+3)(48N^4 + 306N^3 + 715N^2 + 729N + 275), \\ R_N &= 6528N^8 + 54480N^7 + 187144N^6 + 344616N^5 \\ &\quad + 371338N^4 + 239883N^3 + 91396N^2 + 19200N + 1800, \quad \tau_N = \frac{V_N}{R_N}. \end{aligned}$$

Clearly $\tau_N \geq 0$, while $\tau_N \leq 1/16$ follows after clearing the positive denominator from

$$\begin{aligned} R_N - 16V_N &= 3456N^8 + 28752N^7 + 99912N^6 + 191752N^5 + 226106N^4 \\ &\quad + 169691N^3 + 78196N^2 + 19200N + 1800 \geq 0, \end{aligned}$$

where R_N is the denominator displayed in τ_N . Hence $(\tau_N, 0) \in \mathcal{K}$.

Applying (37) to e_3 and cancelling its nonzero scalar factors gives

$$\frac{P_{h+2,3}}{Q_{h+2,3}} = \frac{\ell_P \cdot (1, \mathcal{P}_0 \mathcal{P}_1 \cdots \mathcal{P}_{h-1}(\tau_{h+2}, 0))}{\ell_Q \cdot (1, \mathcal{P}_0 \mathcal{P}_1 \cdots \mathcal{P}_{h-1}(\tau_{h+2}, 0))}.$$

By (31), its chart tends to z_0 ; by (36), the quotient tends to G . Lemma 20 then promotes the third-column limit to all three columns. This proves the main assertion independently.

Formal verification. The accompanying Lean development proves every analytic estimate, polynomial certificate, matrix identity, and limit used in this section. Its final theorem is

`problem25_solved : Problem25Claim.`

It contains no `sorry`. Its axiom audit is

`{propext, Classical.choice, Quot.sound}`.

16 Enhanced numerical verification

- Birkhoff eigenvector $v_+ = (1, r_{21}, r_{31})$ computed at 500-digit precision ($N_{\max} = 350$).
- $L = G$ confirmed to 500 digits.
- Delannoy decomposition: $f(k) > 0$ verified exactly for $k = 0, \dots, 400$ (401 exact rational terms).
- $g(k)/f(k) = G$ to 40+ digits for $k \geq 56$.
- Error ratio $\varepsilon_{k+1}/\varepsilon_k \rightarrow -1/8$ (12 significant figures at $k = 118$).
- k -recurrence operators L_f, L_g verified exactly on all 401 terms; Poincaré polynomial $(\xi - 1)^2(\xi + 1/8)^6$ confirmed algebraically.

17 Status

Problem 2.5 is solved unconditionally.

The proof is the independent construction in Section 15. Its exact ingredients are:

1. the positive adjoint triple moment and its kernel-checked divergence certificate;
2. the rational companion, invariant Miller box, and uniform $1/8$ backward contraction;
3. the fast-solution selection argument and exact initial Catalan pairings;
4. the third-column terminal certificate, followed by the $2/3$ common-projective-limit squeeze.

The accompanying Lean theorem `problem25.solved` : `Problem25Claim` has no `sorry` and uses only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`.

The earlier elliptic/Delannoy sections are retained as independent structural analysis and computational evidence. Additional established results: scalar N -recurrence (order 3, degree pattern $(7, 14, 21, 28)$, Poincaré roots -16 , $-16(17 \pm 12\sqrt{2})$); k -recurrence (order 8, degree 42, Poincaré roots 1 and $-1/8$); Brafman identity with $k(\rho) = 1$; inverse Delannoy closed form.

Note: the irrationality of Catalan’s constant remains open (this is not required by Problem 2.5).

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